Ans. 3 (the ages 16, 17, 16).

4. Review · Quick [NUM] Give the decimal form of $\frac{7}{8}$, then the fraction in lowest terms equal to 0.12.

   Ans. 0.875; $\frac{12}{100} = \frac{3}{25}$.

5. Thread · Formula → table [REG] $f(x) = x^2 - 2x$. Complete a table of values for $x \in \{0\,;\,1\,;\,2\,;\,3\}$.

   Ans. $0\,;\,-1\,;\,0\,;\,3$.

Sheet 55 Thu 25 Mar. 2027 *Critical reading of statements from the media*

1. Warm-up · Quick [PRO] A 10 % increase on 40: what is the final value?

   Ans. $40 \times 1.10 = 44$.

2. Core · Critical thinking [LEC] An article claims: "60 % of accidents involve sober drivers". Can we conclude that driving sober is dangerous?

   Ans. No: the reference population is missing. If the vast majority of drivers are sober, it is normal for them to be the majority among those involved in accidents. Conditional frequencies would have to be compared.

3. Core + · True/false [LEC] True or false? Justify. "Two smokers out of three are men" means that two men out of three smoke.

   Ans. False. This confuses $P_{\text{smoker}}(\text{man})$ and $P_{\text{man}}(\text{smoker})$: the reference populations differ.

4. Review · True/false [STA] True or false? Justify. The median of a data set is always equal to its mean.

   Ans. False. Counter-example: 1, 2, 9: mean 4, median 2.

5. Thread · Reading code [ALG] What is the value of `s` at the end?

```
s = 0
for i in range(1, 5):
    s = s + i
```

   Ans. $s = 1 + 2 + 3 + 4 = 10$.

   Pitfall: `range(1, 5)` stops at 4; 5 is excluded.

Coordinate geometry: vectors and lines, a synthesis

Sheet 56 Tue 30 Mar. 2027 *Vectors and collinearity: consolidation*

1. Warm-up · Quick [GEO] $\vec{u}\begin{pmatrix}3\\-2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-1\\5\end{pmatrix}$. Calculate $\vec{u}+\vec{v}$ and $2\vec{u}$.

   Ans. $\begin{pmatrix}2\\3\end{pmatrix}$; $\begin{pmatrix}6\\-4\end{pmatrix}$.

2. Core · Method [GEO] Are the points $A(-1;2)$, $B(1;3)$ and $C(5;5)$ collinear?

   Ans. $\overrightarrow{AB}\begin{pmatrix}2\\1\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}6\\3\end{pmatrix}$: $2\times 3 - 1\times 6 = 0$: collinear.

3. Core + · Method [GEO] $A(-1;2)$, $B(3;1)$, $C(4;4)$. Find D such that $ABCD$ is a parallelogram.

   Ans. $\overrightarrow{AB} = \overrightarrow{DC}$: $D = A + \overrightarrow{BC} = (-1+1;2+3) = (0;5)$.

4. Review · Quick [PRO] In a class of 24 pupils, 18 have school lunch. What proportion is this, as a percentage?

   Ans. $\frac{18}{24} = 0.75 = 75\,\%$.

5. Thread · Prove [LIT] Show that $(n+1)^2 - (n-1)^2 = 4n$ for every integer n. What can be concluded about the difference of the squares of two consecutive odd numbers (n even)?

   Ans. $(n^2+2n+1) - (n^2-2n+1) = 4n$. For $n = 2k$ even, the difference of the squares of two consecutive odd numbers is $8k$: a multiple of 8.

   The two integers $n-1$ and $n+1$ are odd when n is even.

Sheet 57 Thu 1st Apr. 2027 *Determinant and equations of lines: consolidation*

1. Warm-up · Quick [GEO] What is the gradient of the line $3x + 2y - 6 = 0$?
   Ans. $y = -\frac{3}{2}x + 3$: gradient $-\frac{3}{2}$.
2. Core · Method [GEO] Equation of the line (AB) with $A(1;2)$ and $B(4;8)$.
   Ans. Gradient $\frac{8-2}{4-1} = 2$; $y = 2x + p$, $2 = 2 + p$: $y = 2x$; or $2x - y = 0$.
3. Core + · Method [GEO] Are the lines $2x - y + 1 = 0$ and $4x - 2y + 5 = 0$ parallel, coincident or intersecting?
   Ans. Direction vectors $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$ are collinear: parallel; $(0;1)$ satisfies the first but $4 \times 0 - 2 + 5 = 3 \neq 0$: strictly parallel.
4. Review · Quick [PUI] Calculate $10^{-2} \times 5 \times 10^{4}$.
   Ans. $5 \times 10^{2} = 500$.
5. Thread · Method [EQ] Solve in $\mathbb{R}$: $7 - 2(x - 1) = 3x - 6$.
   Ans. $x = 3$
   Pitfall: $-2(x-1) = -2x - 2$ *(sign error).*

Sheet 58 Mon 5 Apr. 2027 *Plane geometry problems: choosing a method*

1. Warm-up · Quick [GEO] Distance AB for $A(0;0)$ and $B(5;12)$?
   Ans. $\sqrt{25 + 144} = 13$.
2. Core · Method [DEM] $A(0;0)$, $B(4;2)$, $C(-2;4)$. What type of triangle is ABC?
   Ans. $AB^2 = 20$, $AC^2 = 20$, $BC^2 = 36 + 4 = 40 = AB^2 + AC^2$: right-angled isosceles at A.
3. Core + · Method [DEM] Give two methods to prove, using coordinates, that $ABCD$ is a parallelogram.
   Ans. (i) $\overrightarrow{AB} = \overrightarrow{DC}$; (ii) the diagonals $[AC]$ and $[BD]$ have the same midpoint.
4. Review · Reading [LEC] In a class of 30 pupils, a bar chart shows x pupils with the mark 12 and $2x$ pupils with the mark 14; the other marks account for 18 pupils. Find x.
   Ans. $x + 2x + 18 = 30$, so $3x = 12$ and $x = 4$.
5. Thread · Quick [SOL] A pyramid has a square base of side 6 cm and height 5 cm. Calculate its volume.
   Ans. $V = \frac{1}{3} \times 6^2 \times 5 = 60$ cm^3.

Sheet 59 Tue 6 Apr. 2027 *Situations from other subjects*

1. Warm-up · Quick [MOD] A moving object has position $x(t) = 3t + 2$ (in m, t in s). Where is it at $t = 4$?
   Ans. $x(4) = 14$ m.
2. Core · Method [MOD] A second object follows $x_2(t) = 5t - 4$. At what time and where do the objects meet?
   Ans. $3t + 2 = 5t - 4 \Leftrightarrow t = 3$ s; position $x = 11$ m.
3. Core + · Quick [GEO] A velocity vector has coordinates $\begin{pmatrix}3\\4\end{pmatrix}$ (in m/s). What is its magnitude?
   Ans. $\sqrt{3^2 + 4^2} = 5$ m/s.
4. Review · Quick [CAL] Calculate $\frac{1}{2} - \frac{3}{5}$.
   Ans. $\frac{5}{10} - \frac{6}{10} = -\frac{1}{10}$.
5. Thread · Write-up [DEM] ABC is isosceles at A and H is the foot of the altitude from A. What can be stated about H?
   Ans. H is the midpoint of $[BC]$ (the altitude from the apex is also the perpendicular bisector of the base).